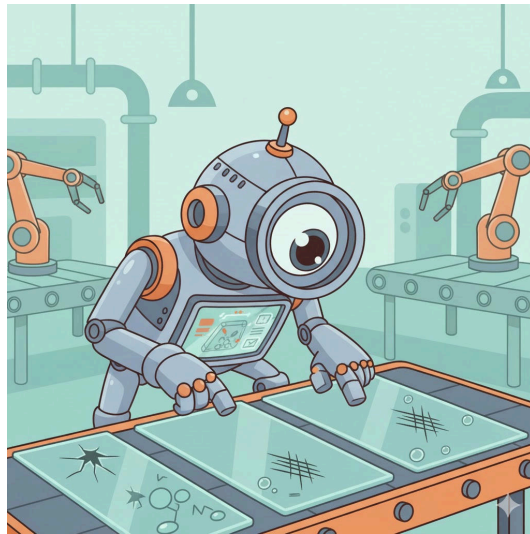


Glass Defect Detection: Evaluating Object Detection Models



Confidential

Projectwork 2

by

Florian Vollmer

Bachelors 2023

Submission date: 2025-12-11

Study Direction: Applied Digital Life Science

Supervisors:

Dr. Stefan Glüge

ZHAW Life Sciences und Facility Management, Wädenswil

Imprint

Recommended Citation:

Florian Vollmer (2025). *Glass Defect Detection: Evaluating Object Detection Models*

.

Keywords: Deep Learning, Object Detectors, Image Processing, Evaluation Metrics

Abstract

Write after work is written

Table of Contents

List of abbreviations	4
1. Introduction	5
1.1. Context and Motivation	5
1.2. Overall Concept	6
1.3. Research Questions	6
1.4. State of the Art	6
2. Methodology	7
2.1. Datasets and Assumptions	7
2.2. Model Selection	7
2.3. Model Finetuning	7
2.4. Model Training	7
2.5. Model Testing	7
2.6. Evaluation Metrics	7
3. Results	8
3.1. Model Results and Statistical Evaluation	8
3.2. Quantitative and Qualitative Evaluation	8
3.3. Evaluation Visualization	8
4. Discussion	9
4.1. Interpretation of Results	9
4.2. Comparison with Expectations	9
5. Conclusion	10
5.1. Summary of Key Findings	10
5.2. Limitations and Challenges	10
5.3. Future Research Directions	10
6. References	11
7. Lists	12
List of Figures	12

List of Tables	12
8. Attachments	13

List of abbreviations

Potential abbreviations

1. Introduction

1.1. Context and Motivation

In the current technological landscape, the implementation of modern technologies is indispensable for ensuring precise and automated processes. Within the industrial sector, quality control represents a critical success factor. Specifically in pharmaceuticals and medical technology, the integrity of the primary packaging material (typically glass containers) is crucial for product safety.

The primary material used for medical glass containers generally consists of borosilicate glass. However, this material can exhibit or contain production-related defects, including particulate inclusions, air bubbles, or flexible fragments known as Lamellae (Foglia, Prete, and Zanda 2015).

The relevance of these quality deficiencies is highlighted by the consequences demonstrated by (Foglia, Prete, and Zanda 2015): *“In recent years, more than 20 product recalls have been associated with glass issues, and over 100 million units of drugs packaged in vials or syringes have been withdrawn from the market.”* This underscores the urgent necessity of reliable detection methods for such flaws.

The rapid advancement of Artificial Intelligence (AI) and fundamental mechanisms like Deep Learning allows for the application of so-called object detectors for the automated recognition of the aforementioned defects. The attractiveness of these AI-based image recognition methods is continuously increasing (Bhatt et al. 2021). However, as these procedures still represent a nascent field of research, comprehensive validation studies on their suitability and the achievability of the necessary performance in industrial use are lacking.

A central challenge in training robust models is the scarcity of extensive datasets (Bhatt et al. 2021). Specifically, the naturally limited availability of data from defective products, caused by the significant imbalance in favor of intact units, complicates the creation of an adequate training dataset (Bhatt et al. 2021).

This paper serves as an exploratory contribution to the evaluation of the current feasibility of AI-supported defect detection. The objective of this work is the comparison of available models on the market and the existence of suitable datasets, in order to provide a well-founded statement on the realisability of precise defect determination.

1.2. Overall Concept

1.3. Research Questions

1.4. State of the Art

2. Methodology

2.1. Datasets and Assumptions

2.2. Model Selection

2.3. Model Finetuning

2.4. Model Training

2.5. Model Testing

2.6. Evaluation Metrics

3. Results

3.1. Model Results and Statistical Evaluation

3.2. Quantitative and Qualitative Evaluation

3.3. Evaluation Visualization

4. Discussion

4.1. Interpretation of Results

4.2. Comparison with Expectations

5. Conclusion

5.1. Summary of Key Findings

5.2. Limitations and Challenges

5.3. Future Research Directions

6. References

Bhatt, Prahar M., Rishi K. Malhan, Pradeep Rajendran, Brual C. Shah, Shantanu Thakar, Yeo Jung Yoon, and Satyandra K. Gupta. 2021. "Image-Based Surface Defect Detection Using Deep Learning: A Review." *Journal of Computing and Information Science in Engineering* 21 (4): 040801. <https://doi.org/10.1115/1.4049535>.

Foglia, Pierfrancesco, Cosimo Antonio Prete, and Michele Zanda. 2015. "An Inspection System for Pharmaceutical Glass Tubes." *WSEAS TRANSACTIONS on SYSTEMS Archive* 14: 123–36. <https://api.semanticscholar.org/CorpusID:62820567>.

7. Lists

List of Figures

List of Tables

8. Attachments